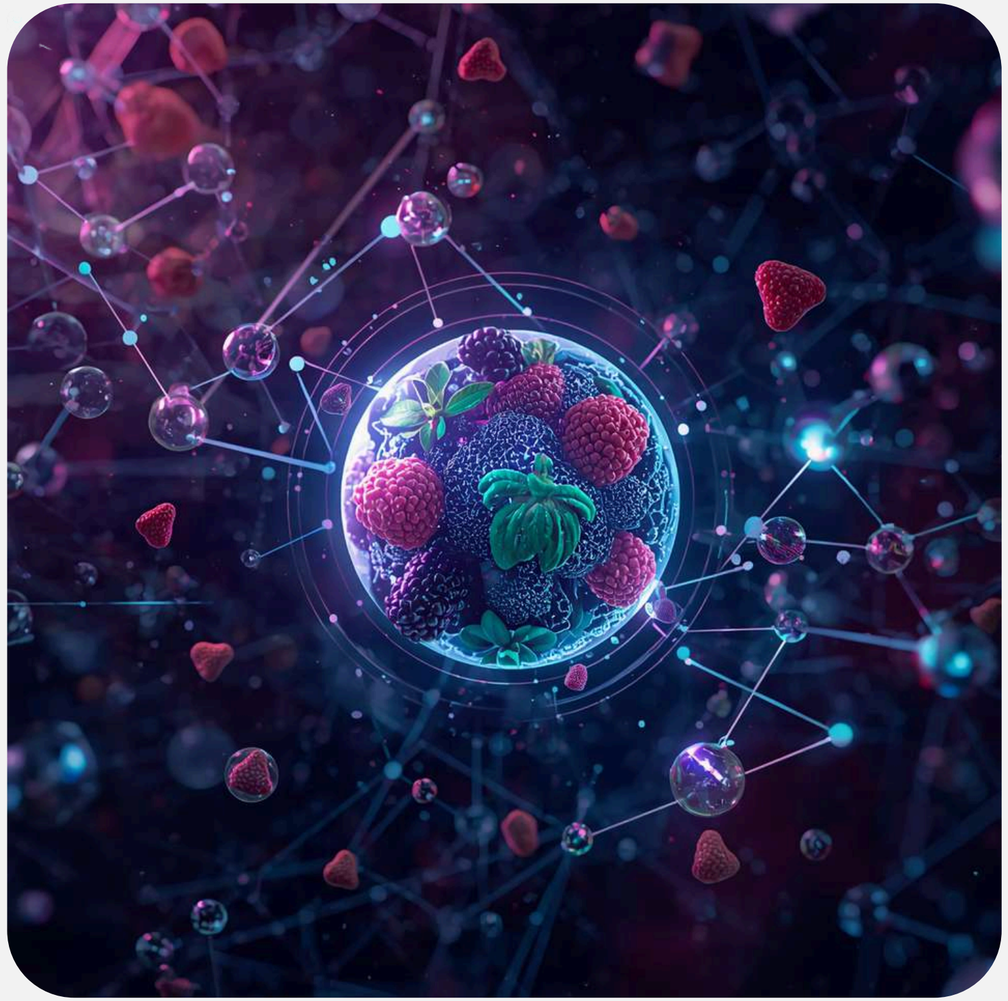
AI-Powered Nutrition Insights

NutriVision App presentation



Team 3 Round 4 : Flutter track

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Project Overview↗

AI-Powered Insights

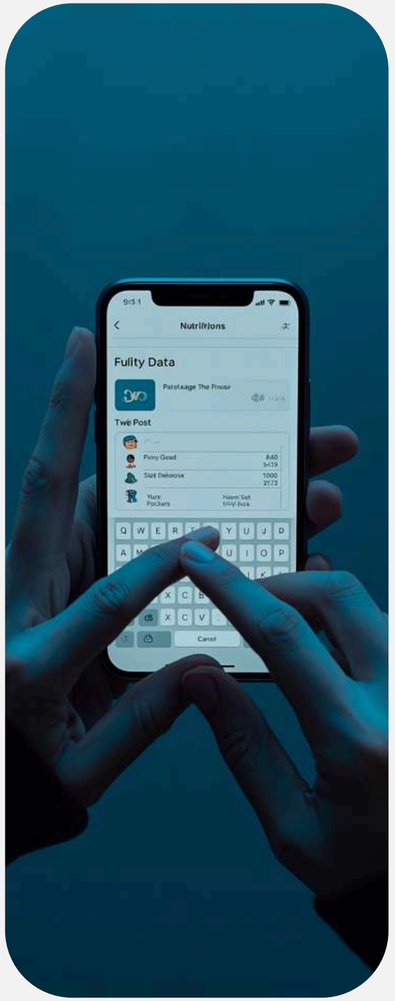
NutriVision delivers personalized dietary insights powered by advanced AI technology for smarter nutrition decisions.

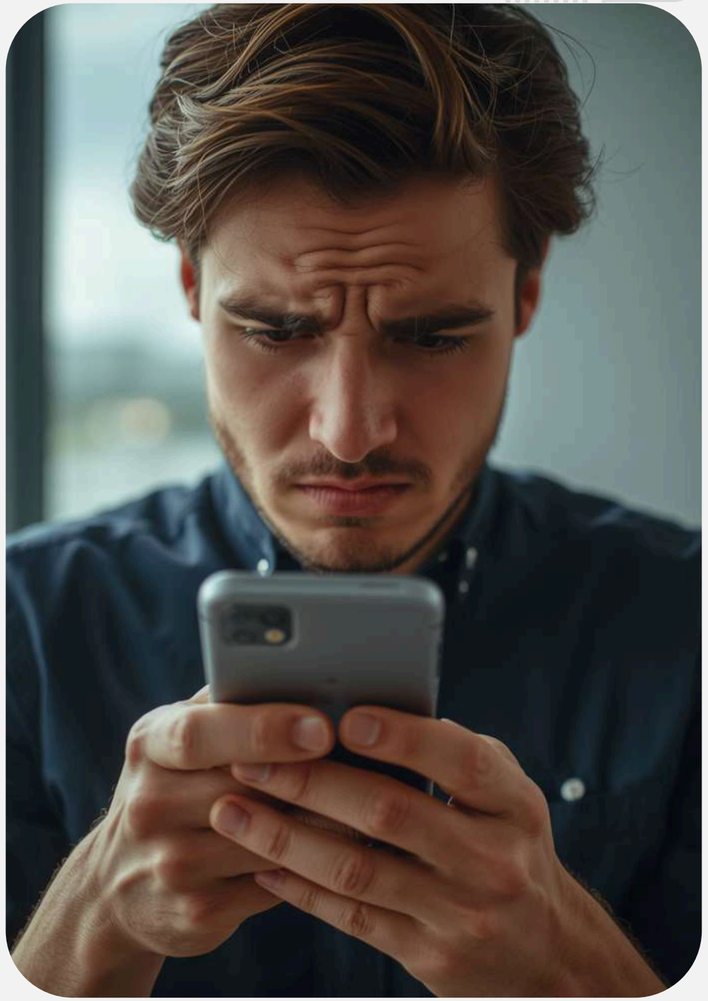
Smart Features

Comprehensive meal logging, detailed nutritional analysis, and intelligent recipe alternatives tailored to your needs.

Cross-Device Sync

Real-time synchronization across all devices with an intuitive, responsive user interface.





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The Problem ↗

Most dietary apps lack personalization, real-time synchronization, and AI-driven insights. Static food databases and manual inputs limit effectiveness, leaving users frustrated with generic recommendations.

Why Current Apps Fall Short ?

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Project Objectives↗

AI Meal Analysis

AI-powered meal description analysis for accurate nutritional breakdown and insights.

Real-Time Sync

Real-time meal log synchronization across all devices using Firebase Firestore.

Recipe Alternatives

Personalized recipe alternatives via advanced prompt engineering and dietary preferences.

Intuitive UI

Responsive Flutter interface with charts and GridView for seamless user experience.

Structured Testing

Comprehensive testing strategy: unit, widget, and integration tests for reliability.

Secure Networking

Secure API communication with Dio interceptors and robust error handling.

Key User Interactions

NutriVision Use Cases Overview

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Log Meal

Capture meals with photos or text descriptions. AI analyzes nutritional content instantly and stores complete history for tracking your dietary journey over time.

05

AI Recipes

Get personalized recipe alternatives powered by AI. Suggestions adapt to your dietary preferences and restrictions, offering healthier options tailored just for you.

05

Sync & Track

Real-time meal log synchronization across all devices. View comprehensive meal history with nutritional charts and manage dietary preferences affecting AI suggestions.

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Functional Requirements ↗

User Account

Complete account management including registration, secure login, and dietary preferences setup.

Nutritional Analysis

Real-time nutritional data retrieval via external API through secure Dio HTTP requests.

Meal Logging

Full CRUD operations for meals with input, edit, delete, and automatic timestamp tracking.

AI Recipe Engine

Intelligent recipe suggestions powered by AI with automatic dietary adaptation and personalization.

Meal History

Comprehensive meal history display with interactive charts and GridView visualization.

Real-Time Sync

Seamless cross-device synchronization powered by Firebase Firestore for instant updates.

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Non-Functional Requirements

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Performance

Lightning-fast nutritional analysis with response times under 2 seconds. Optimized API calls and efficient data caching ensure seamless user experience.

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Security

Robust API call protection with Dio interceptors and Firebase security rules. User data encrypted and access controlled per authenticated user.

09

Reliability

Guaranteed ≥99% uptime with stable real-time sync. Scalable architecture handles growing users while maintaining modular MVC code and accessibility standards.

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MVC Architecture↗

Model

Data management layer handling meals, user profiles, and nutritional information with structured objects for seamless data flow.

View

Flutter UI presentation layer featuring GridView, Stack widgets, and interactive charts for intuitive meal visualization.

Controller

Business logic layer managing API calls, AI prompt generation, and Firebase synchronization for real-time updates.

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System Components↗

User Interface

Flutter-based responsive screens with GridView, Stack, and charts for intuitive meal tracking.

External API

Nutritional analysis service providing comprehensive food data via secure Dio requests.

Data Model

Structured objects for meals, user profiles, and nutritional data with clean architecture.

AI Engine

Recipe alternative generation using prompt engineering tailored to dietary preferences.

Controller Logic

API and AI interaction handling, sync management, and business logic coordination.

Firebase Firestore

Real-time NoSQL database enabling cross-device synchronization and secure data storage.

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User Interface Details↗

Flutter-Powered UI Components

GridView displays meal logs in organized cards. Stack widgets layer nutrition data beautifully. Interactive charts visualize macros and calories. Home, Meal Log, and History screens showcase intuitive navigation patterns.

Responsive Design Excellence

Adaptive layouts ensure seamless experience across all device sizes. From smartphones to tablets, NutriVision maintains visual consistency and intuitive navigation with optimized touch targets.

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Controller Logic & API Integration

Dio Package

Secure, efficient HTTP requests with interceptors for API authentication, error handling, and automatic retries.

AI Prompts

Prompt engineering tailored to dietary restrictions generates personalized recipe alternatives dynamically.

Firebase Sync

Real-time data synchronization via Firebase Firestore ensures seamless cross-device meal log updates.

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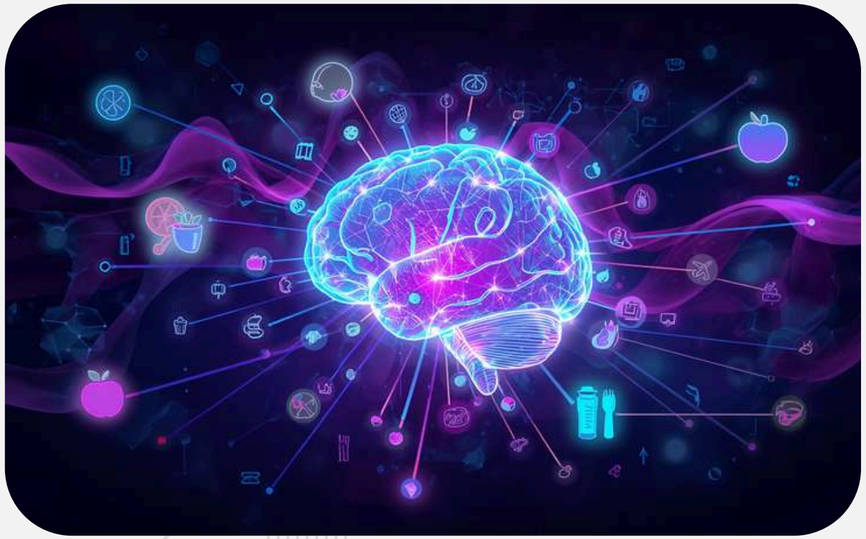
AI Engine Capabilities



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Prompt Engineering

Leverages advanced prompt engineering tailored to individual dietary restrictions including allergies, intolerances, and lifestyle choices like vegan or keto diets.

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Dynamic Recipes

Generates healthier recipe suggestions dynamically based on nutritional analysis, offering alternatives that maintain flavor while improving health outcomes.

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Adaptive Learning

Continuously adapts recommendations based on user preferences, past meal choices, and feedback to deliver increasingly personalized nutrition insights.

NutriVision Database Architecture

Firebase Cloud Firestore ↗

NoSQL Storage

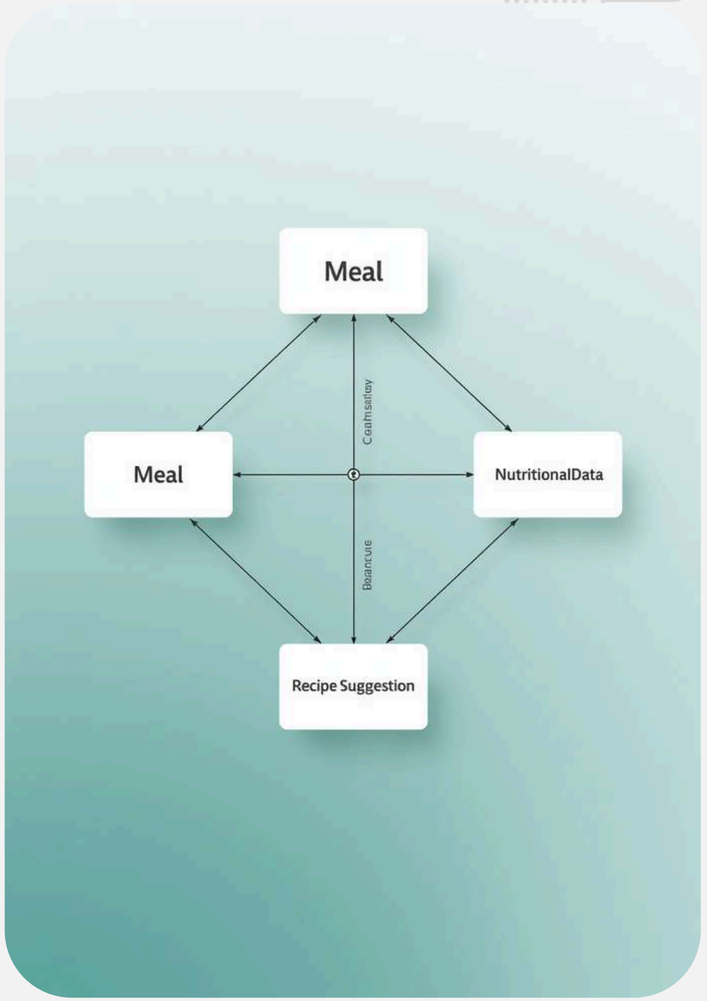
Flexible document store for meals, users, and recipe data with nested collections and scalable architecture.

Real-Time Sync

Instant cross-device synchronization ensures meal logs update seamlessly across all user devices.

Security Rules

Per-user security rules protect data. Indexed queries on user_id, meal_id, and timestamp optimize performance.



Entity-Relationship Diagram

Database Design↗

Four core entities power NutriVision: User (PK: user_id), Meal (PK: meal_id, FK: user_id), NutritionalData (PK: nutrition_id, FK: meal_id), and RecipeSuggestion (PK: suggestion_id, FK: user_id).

Core Data Architecture & Relationships →

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Logical Schema Normalized Tables

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User Table

user_id (PK), name, email, password_hash, dietary_preferences (JSON array storing restrictions, allergies, goals)

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Meal & Nutrition

Meal: meal_id (PK), user_id (FK), description, timestamp.
NutritionalData: nutrition_id (PK), meal_id (FK), calories, protein, carbs, fats, vitamins

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RecipeSuggestion

suggestion_id (PK), user_id (FK), original_meal, suggested_recipe, AI_prompt (stores the engineered prompt used for generation)

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UI/UX Design Principles↗

Simplicity

Minimal information display reduces cognitive load. Clean interfaces guide users naturally through meal logging and nutrition insights.

Consistency

Unified colors, fonts, buttons, and navigation patterns create familiarity. Visual hierarchy emphasizes key actions and data.

Accessibility

Readable fonts, proper contrast ratios, and adequate tap targets ensure usability for all. Responsive feedback confirms user actions.

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Color Palette & Visual Identity

Primary & Secondary Colors

Our color palette is carefully designed to communicate nutritional information intuitively. Each macro color creates instant recognition, while the primary green reinforces our health-focused brand identity across all screens.

Brand Color System

Primary Green (#4CAF50) represents health, vitality, and positive action. Blue indicates protein content, Orange/Yellow for carbohydrates, and Red for fats. Light backgrounds with card layouts and subtle shadows enhance readability.

NutriVision App

Designed Screens Showcase

Splash Screen

Welcome & Branding

Login/Register

User Authentication

Home Dashboard

Daily Overview & Stats

Meal Log

Food Entry & Tracking

Meal Result

Nutritional Analysis

AI Recipe Alternative

Smart Suggestions

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Technology Stack ↗

Flutter & Dart

Cross-platform UI framework with expressive widgets and hot reload for rapid development.

Dio HTTP Client

Powerful HTTP networking with interceptors for secure API calls and error handling.

State Management

Provider and Bloc patterns for scalable, reactive state handling across the app.

Firebase Services

Authentication and Firestore for secure user management and real-time data sync.

AI Engine

OpenAI-powered recipe generation with intelligent prompt engineering.

Cloud Backend

Scalable serverless architecture with Firebase Hosting and cloud functions.

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Development & Deployment ↗

Version Control

Git + GitHub/GitLab for collaborative development, branching strategies, and code review workflows.

CI/CD Pipeline

GitHub Actions and Firebase Hosting pipelines with Flutter Test and Mockito for unit and integration testing.

Deployment

Distribution via Google Play Store, Apple App Store, and Firebase Hosting for web deployment.

Team 3 Round 4

Meet the NutriVision Team

Ahmed Ezz Eldin Elaraby

Team Leader

Omar Shalaby

Developer

Nouran Elfeky

Developer

Omar Abdelmagid

Developer

Jessica Farag

Developer

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AI-powered nutrition insights

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Summary & Next Steps ↗

AI-Driven Insights

NutriVision delivers personalized, AI-powered nutrition insights through intelligent meal analysis and smart recipe alternatives.

Real-Time Sync

Cross-device synchronization via Firebase ensures seamless data access with an intuitive Flutter UI featuring charts and visual feedback.

Next Steps

Moving forward: comprehensive testing phase, deployment to app stores, and continuous user feedback integration for improvement.

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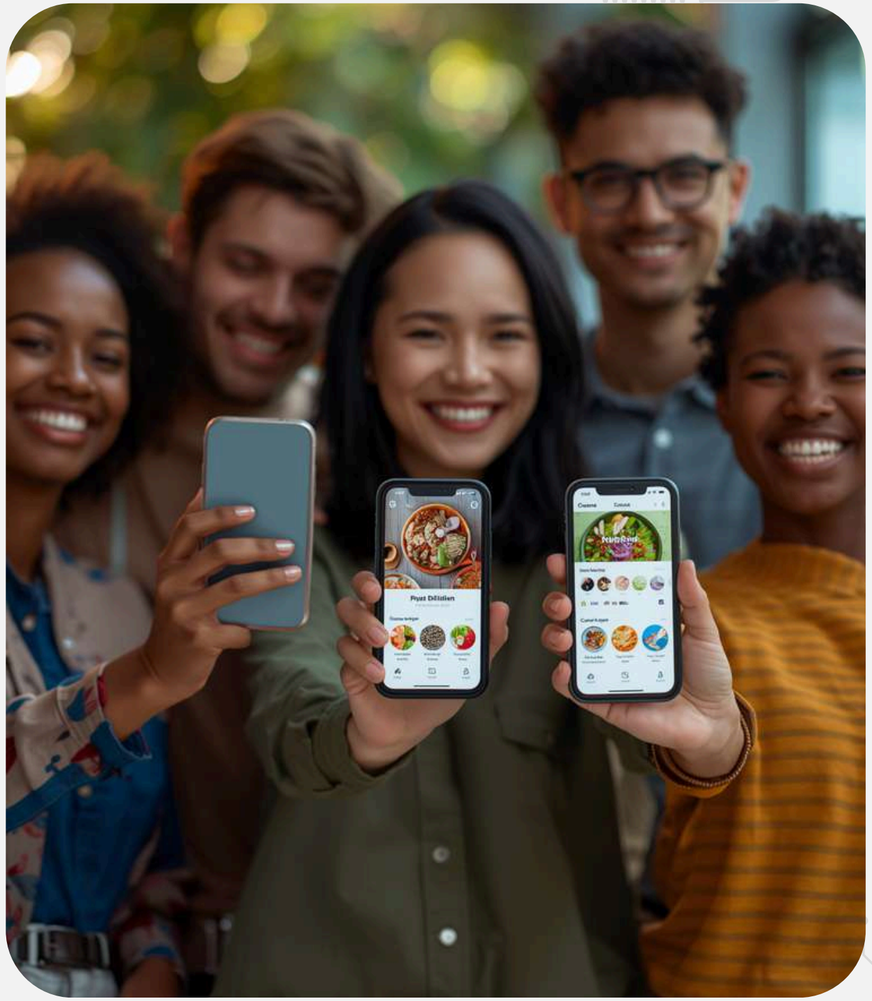
Thank You↗

Thank you for joining us on this journey to revolutionize personalized nutrition through AI-powered insights.